

1 Monotone Strategy Proofs

This appendix provides the detailed proofs regarding monotone calling strategies and their role in equilibrium selection in LCP.

1.1 Monotone Strategies and Weak Dominance

Lemma 1.1. *If a calling strategy violates the first monotonicity condition for a nonzero-measure set of hands for any bet size s , it is weakly dominated. Specifically, if there exists s and measurable sets $A, B \subseteq [0, 1]$ such that:*

1. *The caller calls s with hands in A*
2. *The caller folds s with hands in B*
3. $\sup A \leq \inf B$
4. *A and B have positive measure*

then the strategy is weakly dominated.

Proof. Let σ_C be the non-monotone strategy described above. Since A and B are nonzero-measure, there exist subsets $A' \subseteq A$ and $B' \subseteq B$ such that:

1. A' and B' have positive measure
2. $|A'| = |B'|$

Where $|A|$ and $|B|$ denote the Lebesgue measure of A and B (see Figure 1).

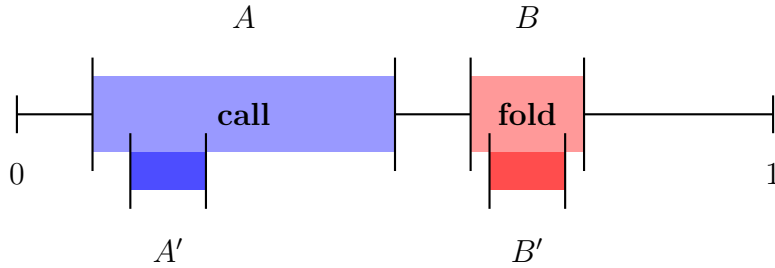


Figure 1: A simple case of sets A and B which violate monotonicity ($\sup A \leq \inf B$). We can find equal-measure subsets $A' \subseteq A$ and $B' \subseteq B$ to swap actions, improving the strategy.

The existence of such subsets follows from a fundamental property of nonatomic measures: since the uniform distribution on $[0, 1]$ is nonatomic (no single point has

positive probability), for any two measurable sets with positive measure, we can always find measurable subsets of equal measure [?]. This property allows us to construct the strategy improvement described below.

Let σ'_C be the strategy which switches the actions for A' and B' , i.e. calls with B' and folds with A' (and behaves identically for all other bet sizes). We now analyze how this change affects the caller's performance against any betting strategy.

Against a bet of size s , the key improvement occurs in two scenarios:

1. When $y \in B'$ and $x \in A'$: σ_C folds while σ'_C calls and wins (since $x \in A$ and $y \in B$ with $\sup A \leq \inf B$)
2. When $y \in A'$ and $x \in B'$: σ_C calls and loses while σ'_C folds (avoiding the loss)

For all other cases, σ_C and σ'_C behave identically, so σ'_C is weakly better than σ_C against every betting strategy.

To show that σ_C is strictly dominated, consider a betting strategy which always bets s . Against this strategy, both scenarios above occur with positive probability (since A' and B' have positive measure), so σ'_C is strictly better than σ_C . Thus, σ_C is weakly dominated. $\square$